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(venv) PS C:\Users\thotl\customer-support-ai> py app.py
Both GOOGLE_API_KEY and GEMINI_API_KEY are set. Using GOOGLE_API_KEY.
Choose an option:
1. Run Demo
2. Interactive Chat
Enter 1 or 2: 1
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ABC TECHNOLOGIES - AI CUSTOMER SUPPORT AUTOMATION DEMO
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--- QUERY 1 ---
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Customer: What are the pricing plans available for your software?
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[Classifier] Analyzing customer query...
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[Classifier] Intent: Sales
```

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[Classifier] Department: Sales
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[Classifier] Requires Approval: False
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```
[Sales] Retrieving knowledge base information...
```

```
[RAG] Sources Retrieved: pricing.txt, pricing_guide.txt
```

```
[Supervisor] Validating and preparing final response...
```

```
[Supervisor] Gemini failed, using fallback response. Error: Error calling model 'gemini-1.5-flash' (UNAUTHENTICATED): 401 UNAUTHENTICATED. {'error': {'code': 401, 'message': 'Request had invalid authentication credentials. Expected OAuth 2 access token, login cookie or other valid authentication credential. See https://developers.google.com/identity/sign-in/web/devconsole-project.'}, 'status': 'UNAUTHENTICATED', 'details': [{'@type': 'type.googleapis.com/google.rpc.ErrorInfo', 'reason': 'ACCESS_TOKEN_TYPE_UNSUPPORTED', 'metadata': {'method': 'google.ai.generativelanguage.v1beta.GenerativeService.GenerateContent', 'service': 'generativelanguage.googleapis.com'}}]}
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```
[Supervisor] Final response generated.
```

```
[Memory] Saving interaction to SQLite...
```

```
[Memory] Interaction saved successfully.
```

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=====
FINAL RESPONSE
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```
Customer, here is the support information I found:
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```
[Source: pricing.txt]
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```
Pricing Guide: Basic plan is $10/month with 5GB storage. Pro plan is $30/month with 50GB storage and priority support. Enterprise pla
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--- QUERY 1 ---
Customer: What are the pricing plans available for your software?

[Classifier] Analyzing customer query...
[Classifier] Intent: Sales
[Classifier] Department: Sales
[Classifier] Requires Approval: False

[Sales] Retrieving knowledge base information...
[RAG] Sources Retrieved: pricing.txt, pricing_guide.txt

[Supervisor] Validating and preparing final response...
[Supervisor] Gemini failed, using fallback response. Error: Error calling model 'gemini-1.5-flash' (UNAUTHENTICATED): 401 UNAUTHENTICATED. {'error': {'code'
: 401, 'message': 'Request had invalid authentication credentials. Expected OAuth 2 access token, login cookie or other valid authentication credential. See
https://developers.google.com/identity/sign-in/web/devconsole-project.', 'status': 'UNAUTHENTICATED', 'details': [{'@type': 'type.googleapis.com/google.rpc
.ErrorInfo', 'reason': 'ACCESS_TOKEN_TYPE_UNSUPPORTED', 'metadata': {'method': 'google.ai.generativelanguage.v1beta.GenerativeService.GenerateContent', 'ser
vice': 'generativelanguage.googleapis.com'}}]}}
[Supervisor] Final response generated.

[Memory] Saving interaction to SQLite...
[Memory] Interaction saved successfully.

=====
FINAL RESPONSE
=====
Customer, here is the support information I found:

[Source: pricing.txt]
Pricing Guide: Basic plan is $10/month with 5GB storage. Pro plan is $30/month with 50GB storage and priority support. Enterprise plan is $100/month with un
limited storage. Annual subscription gets a 20% discount on all plans.

[Source: pricing_guide.txt]
Pricing Guide

1. Basic Plan: $10/month
   - Core business management features
   - Email support

2. Pro Plan: $30/month
   - Advanced analytics
   - Team collaboration tools
   - Priority email support

3. Enterprise Plan: Custom pricing
   - Dedicated account manager
   - Advanced security controls
   - Custom integrations

4. Annual subscriptions receive a 20% discount.

If you need more help, please let us know.
```

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--- QUERY 2 ---
Customer: I forgot my account password.

[Classifier] Analyzing customer query...
[Classifier] Intent: Account
[Classifier] Department: Account
[Classifier] Requires Approval: False

[Account] Retrieving knowledge base information...
[RAG] Sources Retrieved: faq.txt, faq_document.txt

[Supervisor] Validating and preparing final response...
[Supervisor] Gemini failed, using fallback response. Error: Error calling model 'gemini-1.5-flash' (UNAUTHENTICATED): 401 UNAUTHENTICATED. {'error': {'code': 401, 'message': 'Request had invalid authentication credentials. Expected OAuth 2 access token, login cookie or other valid authentication credential. See https://developers.google.com/identity/sign-in/web/devconsole-project.', 'status': 'UNAUTHENTICATED', 'details': [{'@type': 'type.googleapis.com/google.rpc.ErrorInfo', 'reason': 'ACCESS_TOKEN_TYPE_UNSUPPORTED', 'metadata': {'method': 'google.ai.generativelanguage.v1beta.GenerateContent', 'service': 'generativelanguage.googleapis.com'}}]}}
[Supervisor] Final response generated.

[Memory] Saving interaction to SQLite...
[Memory] Interaction saved successfully.

=====
FINAL RESPONSE
=====
Customer, here is the support information I found:

[Source: faq_document.txt]
FAQ Document

1. Password Reset:
  Click "Forgot Password" on the login page and follow the reset instructions.

2. Refund Processing:
  Approved refunds are generally processed within 5-7 business days.

3. Invoice Requests:
  Customers can request invoices from the billing department.

4. Profile Updates:
  Customers can update profile information from the account settings page.

5. Account

[Source: faq.txt]
FAQ: To reset password, click 'Forgot Password' on the login page and follow the email instructions. Refunds are processed within 5-7 business days after approval. To cancel subscription, go to Settings > Billing > Cancel Plan. Account deactivation takes 24 hours to complete.

[Source: faq_document.txt]
ation from the account settings page.

5. Account Activation/Deactivation:
  Contact account support if activation or deactivation assistance is required.

If you need more help, please let us know.
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```

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--- QUERY 3 ---
Customer: My application crashes whenever I upload a file.

[Classifier] Analyzing customer query...
[Classifier] Intent: Technical
[Classifier] Department: Technical Support
[Classifier] Requires Approval: False

[Technical Support] Retrieving knowledge base information...
[RAG] Sources Retrieved: pricing_guide.txt, tech_manual.txt, technical_manual.txt

[Supervisor] Validating and preparing final response...
[Supervisor] Gemini failed, using fallback response. Error: Error calling model 'gemini-1.5-flash' (UNAUTHENTICATED): 401 UNAUTHENTICATED. {'error': {'code': 401, 'message': 'Request had invalid authentication credentials. Expected OAuth 2 access token, login cookie or other valid authentication credential. See https://developers.google.com/identity/sign-in/web/devconsole-project.', 'status': 'UNAUTHENTICATED', 'details': [{'@type': 'type.googleapis.com/google.rpc.ErrorInfo', 'reason': 'ACCESS_TOKEN_TYPE_UNSUPPORTED', 'metadata': {'method': 'google.ai.generativelanguage.v1beta.GenerativeService.GenerateContent', 'service': 'generativelanguage.googleapis.com'}}]}}
[Supervisor] Final response generated.

[Memory] Saving interaction to SQLite...
[Memory] Interaction saved successfully.

=====
FINAL RESPONSE
=====
Customer, here is the support information I found:

[Source: technical_manual.txt]
Technical Manual

1. If the application crashes during file upload:
  - Clear the browser/application cache
  - Confirm the file size is under 50 MB
  - Update the application to the latest version
  - Retry using a supported file format
  - Review system logs for upload-related errors

2. Login problems:
  - Verify username and password
  - Reset password if needed
  - Check whether the a

[Source: tech_manual.txt]
Technical Manual: If the application crashes on file upload, clear the browser cache and ensure file size is under 50MB. For login problems, verify 2FA settings and reset session cookies. For installation issues, ensure OS meets minimum requirements: Windows 10+, macOS 12+, or Ubuntu 20.04+.

[Source: pricing_guide.txt]
Pricing Guide
```

[Supervisor] Final response generated.

[Memory] Saving interaction to SQLite...

[Memory] Interaction saved successfully.

=====

FINAL RESPONSE

=====

Customer, here is the support information I found:

[Source: technical_manual.txt]

Technical Manual

1. If the application crashes during file upload:
 - Clear the browser/application cache
 - Confirm the file size is under 50 MB
 - Update the application to the latest version
 - Retry using a supported file format
 - Review system logs for upload-related errors
2. Login problems:
 - Verify username and password
 - Reset password if needed
 - Check whether the a

[Source: tech_manual.txt]

Technical Manual: If the application crashes on file upload, clear the browser cache and ensure file size is under 50MB. For login problems, verify 2FA settings and reset session cookies. For installation issues, ensure OS meets minimum requirements: Windows 10+, macOS 12+, or Ubuntu 20.04+.

[Source: pricing_guide.txt]

Pricing Guide

1. Basic Plan: \$10/month
 - Core business management features
 - Email support
2. Pro Plan: \$30/month
 - Advanced analytics
 - Team collaboration tools
 - Priority email support
3. Enterprise Plan: Custom pricing
 - Dedicated account manager
 - Advanced security controls
 - Custom integrations
4. Annual subscriptions receive a 20% discount.

If you need more help, please let us know.

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```
--- QUERY 4 ---
Customer: My name is David. I need a refund for my annual subscription.

[Classifier] Analyzing customer query...
[Classifier] Intent: Billing
[Classifier] Department: Billing
[Classifier] Requires Approval: True

[HITL] High-risk request detected.
[HITL] Query: My name is David. I need a refund for my annual subscription.
[HITL] This request requires human supervisor approval.
Supervisor decision (Approve/Reject): approve
[HITL] Request approved by supervisor.

[Billing] Retrieving knowledge base information...
[RAG] Sources Retrieved: faq.txt, faq_document.txt, pricing.txt

[Supervisor] Validating and preparing final response...
[Supervisor] Gemini failed, using fallback response. Error: Error calling model 'gemini-1.5-flash' (UNAUTHENTICATED): 401 UNAUTHENTICATED. {'error': {'code': 401, 'message': 'Request had invalid authentication credentials. Expected OAuth 2 access token, login cookie or other valid authentication credential. See https://developers.google.com/identity/sign-in/web/devconsole-project.'}, 'status': 'UNAUTHENTICATED', 'details': [{'@type': 'type.googleapis.com/google.rpc.ErrorInfo', 'reason': 'ACCESS_TOKEN_TYPE_UNSUPPORTED', 'metadata': {'method': 'google.ai.generativeLanguage.v1beta.GenerativeService.GenerateContent', 'service': 'generativelanguage.googleapis.com'}}]}}
[Supervisor] Final response generated.

[Memory] Saving interaction to SQLite...
[Memory] Interaction saved successfully.

=====
FINAL RESPONSE
=====
David, here is the support information I found:

[Source: faq.txt]
FAQ: To reset password, click 'Forgot Password' on the login page and follow the email instructions. Refunds are processed within 5-7 business days after approval. To cancel subscription, go to Settings > Billing > Cancel Plan. Account deactivation takes 24 hours to complete.

[Source: faq_document.txt]
FAQ Document

1. Password Reset:
   Click "Forgot Password" on the login page and follow the reset instructions.

2. Refund Processing:
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5. Account

[Source: pricing.txt]
Pricing Guide: Basic plan is $10/month with 5GB storage. Pro plan is $30/month with 50GB storage and priority support. Enterprise plan is $100/month with unlimited storage. Annual subscription gets a 20% discount on all plans.

If you need more help, please let us know.
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--- QUERY 5 ---

Customer: What was my previous support issue?

[Classifier] Analyzing customer query...

[Classifier] Intent: Memory

[Classifier] Department: None

[Classifier] Requires Approval: False

[Memory] Retrieving previous customer history from SQLite...

[Memory] Previous Issue: Your previous support issue was: 'My name is David. I need a refund for my annual subscription.' (Intent: Billing, Department: Billing, Time: 2026-06-26T10:28:51.092424).

[Supervisor] Validating and preparing final response...

[Supervisor] Gemini failed, using fallback response. Error: Error calling model 'gemini-1.5-flash' (UNAUTHENTICATED): 401 UNAUTHENTICATED. {'error': {'code': 401, 'message': 'Request had invalid authentication credentials. Expected OAuth 2 access token, login cookie or other valid authentication credential. See <https://developers.google.com/identity/sign-in/web/devconsole-project>.'}, 'status': 'UNAUTHENTICATED', 'details': [{'@type': 'type.googleapis.com/google.rpc.ErrorInfo', 'reason': 'ACCESS_TOKEN_TYPE_UNSUPPORTED', 'metadata': {'method': 'google.ai.generativelanguage.v1beta.GenerateService.GenerateContent', 'service': 'generativelanguage.googleapis.com'}}]}

[Supervisor] Final response generated.

[Memory] Saving interaction to SQLite...

[Memory] Interaction saved successfully.

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FINAL RESPONSE

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David, Your previous support issue was: 'My name is David. I need a refund for my annual subscription.' (Intent: Billing, Department: Billing, Time: 2026-06-26T10:28:51.092424).

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(venv) PS C:\Users\thotl\customer-support-ai> |